IB Biology SL — 2025 Syllabus

THEME B - FORM & FUNCTION

Exam: Paper 1A (MCQ) · Paper 1B (Data) · Paper 2A (Short answer) · Paper 2B (Extended response)

WHAT'S IN THIS GUIDE

B1.1 Carbohydrates & Lipids — B1.2 Proteins — B2.1 Membranes & Transport — B2.2 Organelles — B2.3 Cell Specialization — B3.1 Gas Exchange — B3.2 Transport — B4.1 Adaptation to Environment — B4.2 Ecological Niches

This theme asks: how are biological molecules and cells structured, and how does structure determine function? Biochemistry and cell biology connect directly to physiology.

High-frequency exam topics: protein structure, fluid mosaic model, active/passive transport, stem cells, gas exchange adaptations, cardiac cycle, xylem/phloem, and ecological niches.

B1.1 — Carbohydrates & Lipids

Carbon · Monosaccharides · Polysaccharides · Glycoproteins · Triglycerides · Phospholipids

Carbon forms the backbone of all biological molecules. Its ability to form 4 stable covalent bonds (tetravalency) and bond with other carbon atoms creates the diversity of organic molecules needed for life.

Molecule	Monomer	Bond formed	Structure	Function
Monosaccharide	— (is the monomer)	—	Single ring (glucose: 6C hexose — α or β form). Ribose: 5C pentose.	Immediate energy source (glucose). Component of nucleotides (ribose). Component of polysaccharides.
Disaccharide	Monosaccharides	Glycosidic bond (condensation reaction — water released)	Two monosaccharides joined. Maltose (α-glucose + α-glucose). Sucrose (glucose + fructose). Lactose (glucose + galactose).	Transport form of sugar (sucrose in phloem). Energy storage (short-term).
Starch	α-glucose	α-1,4 glycosidic bonds (amylose: straight chain) + α-1,6 (amylopectin: branched)	Coiled (amylose) or branched (amylopectin). Compact. Insoluble.	Energy storage in PLANTS. Compact — does not affect osmotic potential. Hydrolysed to glucose when needed.
Glycogen	α-glucose	α-1,4 + α-1,6 glycosidic bonds — highly branched	Like starch but MORE branched. Compact. Insoluble.	Energy storage in ANIMALS (liver and muscle). Highly branched = many terminal ends = rapid hydrolysis.
Cellulose	β-glucose	β-1,4 glycosidic bonds — alternating orientation	Long, straight chains (no coiling). H-bonds between parallel chains form microfibrils.	Structural component of PLANT CELL WALLS. Rigidity. Insoluble. Cannot be digested by most animals (no cellulase).
Glycoprotein	Protein + carbohydrate chain	Covalent bond	Protein with oligosaccharide chains attached — on outer surface of cell membrane	Cell recognition (blood group antigens), cell signalling, receptor function, mucus (glycoproteins in mucus membranes)

Lipid type	Composition	Key structural features	Functions
Triglyceride	Glycerol + 3 fatty acids	Ester bonds (condensation × 3). Saturated: no C=C double bonds (solid at room temp). Unsaturated: 1+ C=C double bonds → kink in chain (liquid at room temp, more fluid).	Energy storage (2× more energy per gram than carbohydrates). Insulation (subcutaneous fat). Buoyancy. Protection (organs). Waterproofing (cuticle of leaves).

Phospholipid	Glycerol + 2 fatty acids + phosphate group	Hydrophilic HEAD (phosphate — charged, polar — attracted to water). Hydrophobic TAIL (2 fatty acid chains — non-polar — repelled by water). AMPHIPATHIC molecule (both properties).	Major component of ALL cell membranes. In water, spontaneously form bilayers (tails inward, heads outward) — the basis of the fluid mosaic model.
Cholesterol	Steroid — 4 fused carbon rings + OH group	Non-polar, hydrophobic molecule that intercalates between phospholipids	Membrane fluidity regulation (prevents solidification at low temp; prevents excess fluidity at high temp). Precursor for steroid hormones (testosterone, oestrogen, cortisol), vitamin D, bile salts.

EXAM TIP

α-glucose vs β-glucose: in α-glucose the —OH on carbon 1 is below the ring. In β-glucose it is above. This small structural difference causes enormous functional difference: α links → starch/glycogen (energy storage), β links → cellulose (structural). This frequently appears as a comparison question.

Condensation vs hydrolysis: CONDENSATION joins monomers + releases water. HYDROLYSIS breaks polymers + requires water. Know both terms and which direction each goes.

Saturated = no double bonds = solid at room temp. Unsaturated = double bonds = liquid at room temp. Trans fats are artificial unsaturated fats with double bonds in trans configuration — associated with cardiovascular disease.

B1.2 — Proteins

Amino acids · Peptide bonds · 4 levels of structure · Denaturation · Functions

Proteins are the most functionally diverse macromolecules in biology. Their vast diversity arises from just 20 different amino acids combined in different sequences and folded into specific 3D shapes.

KEY DEFINITIONS

Amino acid: the monomer of proteins. Structure: central carbon (α-carbon) bonded to: (1) amino group (—NH_2), (2) carboxyl group (—COOH), (3) hydrogen (—H), and (4) R group (variable side chain — determines the amino acid's properties and how it interacts with others).

Essential amino acids: amino acids that CANNOT be synthesised by the body and must be obtained from the diet. Humans have 8 (or 9 in some definitions) essential amino acids. Non-essential amino acids can be synthesised from other molecules.

Peptide bond: covalent bond formed between the carboxyl group of one amino acid and the amino group of the next, via condensation (loss of water). —CO—NH— linkage.

Polypeptide: a chain of amino acids linked by peptide bonds. Proteins may consist of one or more polypeptide chains.

Level of protein structure	Description	Bonds maintaining it	Example
Primary structure	The LINEAR SEQUENCE of amino acids in the polypeptide chain — determined by the gene (DNA sequence)	Peptide bonds (covalent — very strong)	Haemoglobin primary structure = specific sequence of ~574 amino acids across 4 chains
Secondary structure	LOCAL folding patterns: α-helix (right-handed coil) and β-pleated sheet (zigzag strands side by side)	Hydrogen bonds between backbone —C=O and —NH groups	Keratin (hair/nails) is mostly α-helix. Silk is mostly β-sheet.
Tertiary structure	The overall 3D SHAPE of a single polypeptide — how the secondary structures fold on each other	Multiple bond types: H-bonds (weak), ionic bonds (charged R groups), disulfide bonds (—S—S— between cysteine R groups — covalent, strong), hydrophobic interactions (non-polar R groups cluster inward away from water)	Globular proteins (e.g. enzymes, haemoglobin, antibodies) have specific tertiary structure critical for function
Quaternary structure	Two or more polypeptide chains (subunits) associated together — ONLY if protein has multiple subunits	Same bonds as tertiary + also between subunits	Haemoglobin: 4 subunits (2α + 2β). Collagen: 3 polypeptide chains. Antibodies: 4 chains.

KEY DEFINITIONS

Denaturation: the disruption of the bonds maintaining secondary, tertiary, and quaternary structure — the protein UNFOLDS and loses its specific 3D shape. Primary structure (peptide bonds) is NOT broken during denaturation.

Causes of denaturation: high temperature (breaks H-bonds and hydrophobic interactions — kinetic energy disrupts bonds), extreme pH (changes ionisation of R groups, disrupts ionic bonds and H-bonds), organic solvents (disrupt hydrophobic interactions), heavy metal ions (disrupt ionic bonds and react with S-H groups).

Consequence: loss of function — enzymes lose catalytic activity, antibodies lose binding specificity, structural proteins lose integrity.

Reversibility: denaturation is often IRREVERSIBLE (especially at very high temperatures). Mild denaturation may be reversible (e.g. pH changes if corrected quickly).

Protein type	Function	Examples
Enzymes	Biological catalysts — lower activation energy, increase reaction rate, highly specific	Amylase, pepsin, DNA polymerase, rubisco
Structural proteins	Provide mechanical support and shape	Collagen (tendons, skin), keratin (hair, nails), actin/myosin (muscle), tubulin (microtubules)
Transport proteins	Carry molecules in blood or across membranes	Haemoglobin (O_2 in blood), channel proteins, carrier proteins (membrane transport)
Signalling/hormones	Chemical messengers	Insulin, glucagon, growth hormone, FSH, LH
Antibodies (immunoglobulins)	Immune defence — bind antigens specifically	IgG, IgM, IgE — specific to each antigen
Receptor proteins	Detect signals and trigger responses	Hormone receptors, neurotransmitter receptors, olfactory receptors

EXAM TIP

4 levels of protein structure — know what holds each together. Primary = peptide bonds. Secondary = H-bonds (backbone). Tertiary = multiple bond types including disulfide bridges. Quaternary = same as tertiary but between subunits.

Denaturation question: always state WHICH bonds are broken and the consequence (loss of 3D shape → loss of function). Note: PRIMARY structure (peptide bonds) is NOT broken by denaturation.

B2.1 — Membranes & Membrane Transport

Fluid mosaic model · Diffusion · Osmosis · Facilitated diffusion · Active transport · Endocytosis

The cell membrane is not just a barrier — it is a dynamic, selectively permeable structure that controls what enters and leaves the cell. The fluid mosaic model describes its structure.

KEY DEFINITIONS

Fluid mosaic model (Singer & Nicolson, 1972): the cell membrane is a FLUID (phospholipids and proteins can move laterally) MOSAIC (proteins embedded like tiles in a mosaic) of a phospholipid bilayer with associated proteins.

Phospholipid bilayer: hydrophilic heads face outward (toward water on both sides), hydrophobic tails face inward (away from water). Forms spontaneously in aqueous environments.

Integral (transmembrane) proteins: span the entire bilayer — involved in transport (channel proteins, carrier proteins) and as receptors.

Peripheral proteins: attached to one surface only — often involved in cell signalling, cytoskeleton anchoring.

Cholesterol: scattered between phospholipids — regulates fluidity (prevents solidification at low temperature; prevents excess fluidity at high temperature).

Glycoproteins: proteins with oligosaccharide chains on the outer surface — cell recognition, communication, immune function.

Transport type	Energy required	Direction	Mechanism	What crosses	Example
Simple diffusion	None (passive)	High → low concentration (down gradient)	Direct diffusion through phospholipid bilayer	Small, non-polar molecules (O_2, CO_2, ethanol, steroid hormones)	O_2 diffuses from alveoli into blood capillaries
Facilitated diffusion	None (passive)	High → low concentration (down gradient)	Via channel proteins (water-filled pores for specific ions/water) or carrier proteins (change shape to carry specific molecules)	Polar molecules, ions, large molecules (glucose, amino acids, ions)	Glucose enters cells via GLUT transporters. Water via aquaporins.
Osmosis	None (passive)	High water potential → low water potential (down water potential gradient)	Water moves through aquaporins (or directly through bilayer) across a semi-permeable membrane	WATER only	Water moves from dilute solution (high Ψ) to concentrated solution (low Ψ)
Active transport	YES — ATP required	Low → high concentration (AGAINST gradient)	Carrier proteins (pumps) use ATP to change shape and move specific molecules against gradient	Specific ions and molecules	Na^+/K^+ ATPase pump — $3Na^+$ out, $2K^+$ in. Mineral ion uptake by root hair cells.
Endocytosis	YES — ATP required	Into cell	Cell membrane engulfs material → membrane folds inward → forms vesicle inside cell	Large particles, macromolecules, pathogens	Phagocytosis (solids — white blood cells engulfing bacteria). Pinocytosis (liquids/dissolved substances).
Exocytosis	YES — ATP required	Out of cell	Vesicle fuses with cell membrane → contents released outside cell	Large molecules, cell products	Insulin secretion. Neurotransmitter release. Enzyme secretion.

⚠ WATCH OUT / COMMON MISTAKE

Osmosis is NOT 'water moving from low to high concentration'. Osmosis is 'water moving from HIGH water potential (low solute concentration) to LOW water potential (high solute concentration)'. Water moves toward MORE concentrated solutions — this is counterintuitive. Think of it as water moving from where it is more abundant (dilute solution) to where it is less abundant (concentrated solution).

Active transport moves molecules AGAINST the concentration gradient. It ALWAYS requires ATP. Facilitated diffusion is PASSIVE — does not require ATP even though it uses proteins.

Factors affecting membrane permeability: temperature (higher temp → more fluid → more permeable, eventually denatures proteins → very permeable), organic solvents (dissolve lipids → disrupt bilayer).

B2.2 — Organelles & Compartmentalization

Why compartmentalize · Organelle functions · Protein secretion pathway

Compartmentalization allows different chemical reactions to occur simultaneously without interfering with each other. Each organelle maintains a specific internal environment (pH, ion concentration, enzyme content) optimised for its function.

PROCESS: Protein Secretion Pathway (Endomembrane System)

1. SYNTHESIS: Ribosomes on rough ER synthesise proteins destined for secretion or the membrane.
2. INITIAL MODIFICATION: Proteins enter the lumen of the rough ER — disulfide bonds form, initial folding occurs, signal peptide removed.
3. TRANSPORT: Small vesicles bud off from rough ER and carry proteins to the Golgi apparatus.
4. GOLGI PROCESSING: Golgi apparatus receives, modifies (adds carbohydrate chains → glycoprotein), sorts, and packages proteins.
5. EXPORT: Vesicles bud off from Golgi — some fuse with lysosomes (hydrolytic enzymes), others travel to cell membrane.
6. SECRETION: Secretory vesicles fuse with cell membrane → exocytosis → proteins released outside cell.
7. Example: insulin synthesis in pancreatic β cells follows this exact pathway.

EXAM TIP

The secretion pathway order: Ribosome (on RER) → Rough ER → Transport vesicle → Golgi → Secretory vesicle → Cell membrane (exocytosis). Know this sequence — 'order of organelles in protein secretion' is a classic exam question.

Compartmentalization advantages: (1) keeps reactions separate that would interfere, (2) creates concentration gradients for driving reactions (e.g. H^+ gradient in mitochondria), (3) localises hydrolytic enzymes safely (lysosomes), (4) creates different pH environments for different enzymes.

B2.3 — Cell Specialization

Differentiation · Stem cells · Potency · SA:V ratio · Specialized cells

KEY DEFINITIONS

Cell differentiation: the process by which a cell develops a more specialised function by expressing a subset of its genes. ALL cells in an organism have the same DNA, but different genes are expressed (switched on) in different cell types.

Stem cell: an undifferentiated cell capable of: (1) SELF-RENEWAL — dividing to produce more stem cells, and (2) DIFFERENTIATION — developing into more specialised cell types.

Totipotent: can differentiate into ANY cell type, including extra-embryonic tissue (placenta). Only the zygote and very early embryo cells (first few divisions). Most powerful.

Pluripotent: can differentiate into any cell type of the body but NOT extra-embryonic tissue. Embryonic stem cells from the inner cell mass of the blastocyst (5–6 days after fertilisation). Used in research and potential therapies.

Multipotent: can differentiate into a limited range of RELATED cell types. Adult stem cells (e.g. haematopoietic stem cells in bone marrow → all blood cell types; neural stem cells). Less versatile but ethically less controversial.

Induced pluripotent stem cells (iPSCs): adult cells reprogrammed to a pluripotent state by introducing specific transcription factors (Yamanaka factors). Avoid ethical issues of embryonic stem cells.

Surface Area to Volume Ratio (SA:V)

As a cell grows larger, its VOLUME increases faster than its SURFACE AREA. This is because volume scales with the cube of radius (r^3) while surface area scales with the square (r^2).

Example: a cell with radius 1μm has SA=4π, V=(4/3)π, SA:V=3. A cell with radius 2μm has SA=16π, V=(32/3)π, SA:V=1.5. HALVED as cell doubles in size.

WHY SA:V matters: nutrients enter and waste leaves through the surface (membrane). As the cell grows, the volume of cytoplasm needing supply increases faster than the surface available to supply it → diffusion distances become too long → cell cannot function efficiently.

This limits cell size — cells must divide (mitosis) before they get too large. It also explains why many cells have adaptations to increase effective SA: microvilli in intestinal epithelium, flat shape of red blood cells, branching of neurons.

SA:V ratio is also relevant to thermoregulation — small animals have high SA:V → lose heat faster → need higher metabolic rate.

Specialised cell	Key structural adaptations	Function these adaptations serve
Red blood cell (erythrocyte)	Biconcave disc — increases SA for gas exchange. No nucleus — maximum space for haemoglobin. No mitochondria — won't consume O_2 it carries. Flexible membrane — can squeeze through capillaries.	Carry O_2 (bound to haemoglobin) from lungs to tissues. CO_2 transport back to lungs.
Sperm cell	Acrosome (head) contains hydrolytic enzymes to penetrate egg. Mitochondria in midpiece — provide ATP for flagellum. Long flagellum — motility. Haploid nucleus.	Deliver haploid genetic material to egg for fertilisation.
Muscle cell (myocyte)	Many mitochondria (ATP for contraction). Many myofibrils (contractile proteins actin and myosin). SR (Ca^{2+} storage). Multiple nuclei (muscle fibres are syncytia).	Contraction — produce movement.
Neuron (nerve cell)	Very long axon (1m+) — fast signal transmission over distance. Myelin sheath (Schwann cells) — saltatory conduction, insulation. Many dendrites — receive multiple inputs. Synaptic knobs — release neurotransmitters.	Rapid transmission of electrical signals over long distances.
Root hair cell (plant)	Long hair-like projection — greatly increases SA. No chloroplasts (underground — no light). Large vacuole.	Absorption of water (osmosis) and mineral ions (active transport) from soil.

EXAM TIP

Stem cell exam questions: ALWAYS state the POTENCY level and its meaning. 'Pluripotent' is not sufficient — explain 'can differentiate into any cell type of the body but not extra-embryonic tissue'.

Differentiation: all cells have the same DNA but express different genes. This is controlled by transcription factors. Differentiation is generally irreversible in normal development.

B3.1 — Gas Exchange

Lungs · Alveoli · Ventilation · Leaf gas exchange · Stomata · Fick's law

KEY DEFINITIONS

Gas exchange: the process of taking up O_2 and releasing CO_2 across a respiratory surface. Requires: large surface area, thin exchange surface, moist surface, maintenance of concentration gradient.

Fick's law of diffusion: Rate of diffusion ∝ (Surface area × Concentration difference) / Thickness of exchange surface. Organisms maximise SA and concentration difference, minimise thickness.

Alveolus (plural alveoli): tiny air sacs in the lungs — the gas exchange surface. Adaptations: enormous SA (~70m² in both lungs — size of a tennis court), one-cell-thick wall (type I pneumocytes), rich capillary network (concentration gradient maintained by blood flow), moist lining (dissolves gases).

Tidal volume: volume of air inhaled/exhaled in one normal breath (~500mL at rest).

Vital capacity: maximum volume of air that can be exhaled after maximum inhalation.

Residual volume: volume of air remaining in lungs after maximum exhalation — prevents alveoli from collapsing.

PROCESS: Ventilation — Inspiration and Expiration

1. INSPIRATION (breathing in — ACTIVE): Diaphragm contracts (flattens) + external intercostal muscles contract (ribs move up and out) → thoracic volume INCREASES → pressure in lungs DECREASES below atmospheric → air flows IN.
2. EXPIRATION (breathing out — PASSIVE at rest): Diaphragm and external intercostals RELAX → elastic recoil of lungs → thoracic volume DECREASES → pressure in lungs INCREASES above atmospheric → air flows OUT.
3. Forced expiration (active): Internal intercostal muscles contract + abdominal muscles contract → further decrease in thoracic volume → more air expelled.
4. Gas exchange at alveoli: O_2 diffuses from alveolar air (high pO_2) into blood (low pO_2). CO_2 diffuses from blood (high pCO_2) into alveolar air (low pCO_2). Both down their concentration gradients.

Gas Exchange in Leaves

Structure	Location	Function in gas exchange
Stomata	Mainly lower epidermis (abaxial surface) — fewer on upper surface in most plants	Pores for gas exchange: CO_2 enters (for photosynthesis), O_2 exits (waste of photosynthesis), water vapour exits (transpiration)
Guard cells	Pair of cells flanking each stoma	Regulate stomatal opening by changing turgor. Open in light (K^+ pumped in → water follows by osmosis → cells swell and bow open). Close in dark or drought.
Spongy mesophyll	Middle layer of leaf — loose cells with air spaces	Large intercellular air spaces → large internal SA for gas exchange. Cells moist — gases dissolve before diffusing into cells.
Palisade mesophyll	Upper layer — densely packed cells with many chloroplasts	Main site of photosynthesis — high demand for CO_2, produces O_2

Stomatal Adaptations in Xerophytes (Drought-adapted plants)

Xerophytes face the challenge of minimising water loss (transpiration) while still exchanging gases for photosynthesis. They have evolved multiple adaptations:

Fewer stomata — reduces total area for water loss. Sunken stomata — positioned in pits below the leaf surface, where humid air accumulates, reducing the concentration gradient for water vapour.

Hairy (pubescent) leaves — leaf hairs trap a layer of humid air near the surface, reducing the water vapour gradient and slowing transpiration.

Thick waxy cuticle — reduces direct evaporation from leaf surface (not through stomata).

CAM metabolism (Crassulacean Acid Metabolism) — stomata open at NIGHT to take in CO_2 (stored as malic acid), closed during the day when temperatures are highest. Reduces daytime water loss.

Examples: cacti, agaves, aloe vera — all adapted to water-scarce environments.

EXAM TIP

Fick's law: Rate of diffusion ∝ (SA × concentration difference) / thickness. Applied to alveoli: great SA (many alveoli), steep gradient (blood flow removes O_2 and CO_2 rapidly), thin wall (one cell thick). State all three factors when explaining gas exchange efficiency.

Guard cell mechanism: in LIGHT → guard cells pump K^+ ions in by active transport → water potential decreases → water enters by osmosis → guard cells become turgid → stoma opens. In DARK/drought → opposite.

B3.2 — Transport

Blood vessels · Cardiac cycle · Coronary disease · Xylem/Phloem · Transpiration · Roots

Blood vessel	Wall structure	Key features	Function
Artery	Thick wall: elastic fibres + thick smooth muscle layer + thin endothelium	No valves. Carries blood at HIGH pressure. Pulse felt. Lumen relatively small.	Carry oxygenated blood AWAY from heart (except pulmonary artery). Withstand and maintain high pressure.
Vein	Thin wall: thin muscle + elastic + endothelium	VALVES present (prevent backflow). Carries blood at LOW pressure. Large lumen. Blood helped by skeletal muscle action.	Carry deoxygenated blood TOWARD heart (except pulmonary vein). Return blood under low pressure.
Capillary	Wall = ONE cell thick (single layer of endothelium)	No muscle or elastic. Very small diameter. Huge total SA. Permeable to small molecules.	EXCHANGE — O_2, glucose, CO_2, waste cross by diffusion. Form capillary beds in every tissue.

PROCESS: Cardiac Cycle — One Heartbeat

1. ATRIAL SYSTOLE: Atria contract → AV valves (tricuspid and bicuspid/mitral) open → blood pushed from atria into ventricles. Ventricles relaxed.
2. VENTRICULAR SYSTOLE: Ventricles contract → pressure rises → AV valves CLOSE (prevents backflow to atria — 'lub' sound) → semilunar valves (pulmonary and aortic) OPEN → blood ejected into pulmonary artery and aorta.
3. DIASTOLE: All chambers relax → pressure falls → semilunar valves CLOSE (prevents backflow — 'dub' sound) → AV valves open → blood flows from veins into atria and passively into ventricles → cycle repeats.
4. Coronary arteries supply the heart muscle itself with blood. Coronary artery disease: atherosclerosis (plaques of cholesterol + fibrous tissue narrow lumen) → reduced blood flow → angina/heart attack (myocardial infarction).

Plant Transport — Xylem and Phloem

Feature	Xylem	Phloem
Contents transported	Water and dissolved mineral ions (inorganic)	Sucrose (and other organic solutes), amino acids — BOTH directions
Direction of transport	One direction only — ROOT to LEAF (upward)	Both directions — from SOURCE (leaf) to SINK (roots, fruit, growing tips)
Cells	Dead cells — hollow tubes. Walls thickened with LIGNIN (waterproof, rigid, gives strength).	Living cells — sieve tube elements (no nucleus, few organelles) + companion cells (full organelles, control sieve tube metabolism)
Driving force	Cohesion-tension: transpiration pulls water column from above. Root pressure from below (minor). Adhesion to xylem walls assists.	Active loading of sucrose into phloem at source → decreases water potential → water enters by osmosis → creates pressure → sucrose pushed toward sinks (mass flow hypothesis)

Energy requirement	Passive — transpiration drives flow without direct ATP input to xylem	Requires ATP for active loading of sucrose into sieve tubes at the source

KEY DEFINITIONS

Transpiration: the loss of water vapour from plant leaves (mainly through stomata). Drives the transpiration stream in xylem.

Cohesion-tension mechanism: water evaporates from mesophyll cells → water drawn from xylem into mesophyll → tension (negative pressure) develops in xylem → cohesion between water molecules (H-bonds) means the water column 'pulls' upward → water drawn up from roots.

Mycorrhizae: mutualistic fungi associated with plant roots — fungal hyphae extend far into soil, greatly increasing absorptive surface area for water and phosphate uptake. Plant provides sugars in return.

Root hair cells: long projections of root epidermal cells that greatly increase SA for absorption. Mineral ions absorbed by active transport. Water absorbed by osmosis (soil water is more dilute than root cell cytoplasm).

EXAM TIP

Cardiac cycle sound: VALVES closing make the sounds. 'Lub' = AV valves closing (start of ventricular systole). 'Dub' = semilunar valves closing (start of diastole). Valves OPEN silently.

Xylem vs phloem: the most examined plant transport comparison. Key differences: dead vs living cells, one-way vs two-way flow, water vs sucrose, lignin vs sieve plates.

B4.1 — Adaptation to Environment

Abiotic factors · Biomes · Coral reefs · Desert & Rainforest adaptations · Thermoregulation · Extremophiles

Biome	Key abiotic conditions	Representative adaptations	Examples of organisms
Tropical rainforest	High rainfall (>200cm/year), constant high temperature (~26°C), high humidity, intense light at canopy — low light at forest floor	Drip tips (rapid water runoff), buttress roots (support in shallow soils), epiphytes (grow on trees to reach light), large leaves (maximise light capture in understory), thick bark	Jaguars, toucans, bromeliads, lianas, kapok trees
Desert	Very low rainfall (<25cm/year), extreme temperature variation (day/night), high light intensity, low humidity	CAM photosynthesis (stomata open at night), water storage in stems (cacti — succulent), reduced/absent leaves (spines — reduce SA and water loss), deep tap roots OR shallow widespread roots, waxy cuticle, nocturnal behaviour, concentrated urine, counter-current heat exchange	Cacti, camels, kangaroo rats, fennec foxes, Gila monsters
Tundra	Very low temperature, permafrost, low rainfall, short growing season, strong winds	Low-growing cushion plants (reduce wind exposure), dark pigments (absorb heat), antifreeze proteins in cold-adapted fish, fat stores for insulation, hibernation	Polar bears, Arctic foxes, lemmings, Arctic mosses, lichens
Coral reef	Warm (26–29°C), shallow, clear, salty, low nutrient water, high light penetration	Zooxanthellae symbiosis (photosynthetic algae in coral polyp tissue), calcium carbonate skeleton, massive biodiversity (many niches), camouflage, territorial behaviour	Coral polyps, clownfish, sea turtles, parrotfish, moray eels

KEY DEFINITIONS

Abiotic factors: non-living physical and chemical conditions of the environment — temperature, light intensity, water availability, humidity, pH, salinity, dissolved oxygen, soil composition.

Ectotherm: organism whose body temperature is determined by the external environment (reptiles, fish, amphibians, invertebrates). Cannot generate significant metabolic heat. Behaviour is crucial for thermoregulation (basking, shade-seeking).

Endotherm: organism that generates its own body heat through metabolism (birds and mammals). Maintain a relatively constant core body temperature regardless of external conditions. Higher metabolic rate required.

Coral reef: built by coral polyps — tiny animals that secrete calcium carbonate ($CaCO_3$) skeletons. Zooxanthellae (symbiotic photosynthetic dinoflagellates) live within coral tissue, providing photosynthates to coral. Coral bleaching: when water temperatures rise even slightly (1–2°C), zooxanthellae are expelled, coral turns white, and dies if stress continues.

Extremophiles: organisms that thrive in extreme conditions. Thermophiles (hot environments — Taq polymerase from Thermus aquaticus). Halophiles (high salt — salt flats). Acidophiles (low pH). Psychrophiles (very cold — Antarctic ice). Significance: Taq polymerase from thermophiles revolutionised PCR.

EXAM TIP

Desert vs rainforest adaptation questions are very common in Paper 2B. Structure your answer: name the adaptation → explain the mechanism → link to the specific abiotic challenge it addresses.

Thermoregulation: ectotherm = temperature FROM environment. Endotherm = temperature FROM metabolism. Endotherms have much higher food requirements. Both use BEHAVIOUR for thermoregulation.

B4.2 — Ecological Niches

Fundamental vs realized niche · Types of nutrition · Adaptations for feeding · Competitive exclusion

KEY DEFINITIONS

Ecological niche: the functional role of a species in its ecosystem — includes what it eats, what eats it, where it lives, when it is active, its effect on the environment, and all abiotic factors it can tolerate. A species' 'address AND profession' in the ecosystem.

Fundamental niche: the full range of conditions and resources a species could theoretically use — if it were not limited by other species. Determined by the species' physiology and behaviour.

Realized niche: the actual range of conditions and resources a species DOES use — narrowed by competition, predation, and parasitism from other species. Always smaller than or equal to the fundamental niche.

Competitive exclusion principle (Gause's law): two species with IDENTICAL niches cannot coexist indefinitely — one will outcompete and exclude the other. Long-term coexistence requires some niche differentiation.

Resource partitioning: closely related species coexist by dividing up resources (food, habitat, time of activity) — reducing competition. Each occupies a slightly different realized niche.

Nutritional type	Energy source	Carbon source	Examples	Key features
Photoautotroph	Light (photosynthesis)	CO_2 (inorganic)	Plants, algae, cyanobacteria	Produce own organic molecules from inorganic sources. Foundation of most food chains. Produce O_2.
Chemoautotroph	Chemical oxidation (chemosynthesis)	CO_2 (inorganic)	Nitrosomonas (oxidises NH_4^+), sulphur bacteria in hydrothermal vents	Important in nutrient cycling (nitrification). Basis of deep-sea hydrothermal vent communities (no sunlight).
Holozoic heterotroph	Organic molecules consumed	Organic molecules consumed	Animals — ingest, digest, absorb, assimilate food	Complex digestive systems. Herbivore, omnivore, carnivore sub-types based on diet.
Saprotrophic heterotroph	Organic molecules from decomposing material	Organic molecules from decomposing material	Fungi, many bacteria	Secrete enzymes externally (extracellular digestion) → absorb products. Decomposers — recycle nutrients.
Parasitic heterotroph	Organic molecules from living host	Organic molecules from living host	Tapeworms, fleas, mistletoe, Plasmodium (malaria)	Benefit at host's expense. May not kill host immediately. Often reduced digestive systems (absorb digested food from host).

EXAM TIP

Fundamental vs realized niche: ALWAYS distinguish these. Fundamental = what COULD be used. Realized = what IS actually used (reduced by interspecific competition). Use the Gause competitive exclusion principle to explain why realized < fundamental.

Saprotrophic nutrition is often confused with holozoic — the key difference is EXTERNAL digestion (saprotrophic) vs internal digestion (holozoic). Fungi do not ingest food — they secrete enzymes and absorb the products.

LINKS TO OTHER TOPICS

B1.1/B1.2 Biochemistry → C1.1 (enzymes work on these molecules), C1.2/C1.3 (glucose is the substrate for respiration and photosynthesis)

B2.1 Membrane transport → D2.3 (water potential), C1.2 (ATP powers active transport), C2.2 (ion channels in neurons)

B2.3 Stem cells → D2.1 (mitosis produces stem cells), D3.1 (embryonic development)

B3.1 Gas exchange → C1.2 (O_2 used in aerobic respiration), C1.3 (CO_2 used in photosynthesis)

B3.2 Transport → C3.1 (integration of cardiovascular and respiratory systems), D3.3 (insulin transported in blood)

B4.1/B4.2 Ecology → C4.1 (populations and communities), C4.2 (energy transfer), D4.2/D4.3 (ecosystem change)

THEME B — EXAM READINESS CHECKLIST

B1.1: **CARBOHYDRATES & LIPIDS**
- ☑ α vs β glucose and consequences
- ☑ Condensation vs hydrolysis
- ☑ Starch vs glycogen vs cellulose (structure AND function)
- ☑ Phospholipid structure and why it forms bilayer
- ☑ Saturated vs unsaturated fatty acids

B1.2: **PROTEINS**
- ☑ Amino acid structure (4 components)
- ☑ 4 levels of protein structure and bonds
- ☑ Denaturation — causes and what is/isn't broken
- ☑ Functions of proteins

B2.1: **MEMBRANES & MEMBRANE TRANSPORT**
- ☑ Fluid mosaic model — all components
- ☑ 6 types of membrane transport — direction, energy, mechanism
- ☑ Osmosis direction (water potential)
- ☑ Endo/exocytosis

B2.2: **ORGANELLES & COMPARTMENTALISATION**
- ☑ Protein secretion pathway in order
- ☑ Why compartmentalization is advantageous
- ☑ Each organelle structure-function link

B2.3: **CELL SPECIALISATION**
- ☑ Totipotent vs pluripotent vs multipotent
- ☑ SA:V ratio — calculate and explain implications
- ☑ 3 specialised cells — structural adaptations linked to function

B3.1: **GAS EXCHANGE**
- ☑ Fick's law applied to alveoli
- ☑ Inspiration vs expiration mechanism
- ☑ Guard cell mechanism
- ☑ Xerophyte adaptations — 4+

B3.2: **TRANSPORT SYSTEMS**
- ☑ 3 blood vessel types — structure AND function
- ☑ Cardiac cycle 3 stages
- ☑ Xylem vs phloem — all features
- ☑ Cohesion-tension mechanism

B4.1: **ADAPTATION TO ENVIRONMENT**
- ☑ Desert AND rainforest adaptations (3+ each)
- ☑ Ectotherm vs endotherm
- ☑ Coral reef — zooxanthellae relationship
- ☑ Coral bleaching mechanism

B4.2: **ECOLOGICAL NICHES**
- ☑ Fundamental vs realized niche
- ☑ Competitive exclusion principle
- ☑ 5 nutritional types — energy source, carbon source, example